

```
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
```

```
#define MAXPAROLA 30
#define MAXRIGA 80
```

```
int main(int argc, char *argv[])
{
    int freq[MAXPAROLA]; /* vettore di contatori
delle frequenze delle lunghezze delle parole */
    char riga[MAXRIGA];
    int i, inizio, lunghezza;
    FILE *f;
```

```
for(i=0; i<MAXPAROLA; i++)
    freq[i]=0;
```

```
if(argc != 2)
```

```
{
    printf(stderr, "ERRORE, serve un parametro con il nome del file\n");
    exit(1);
}
```

```
f = fopen(argv[1], "r");
if(f==NULL)
```

```
{
    printf(stderr, "ERRORE, impossibile aprire il file %s\n", argv[1]);
    exit(1);
}
```

```
while( fgets( riga, MAXRIGA, f ) != NULL )
```



Advanced IO

System Input/Output

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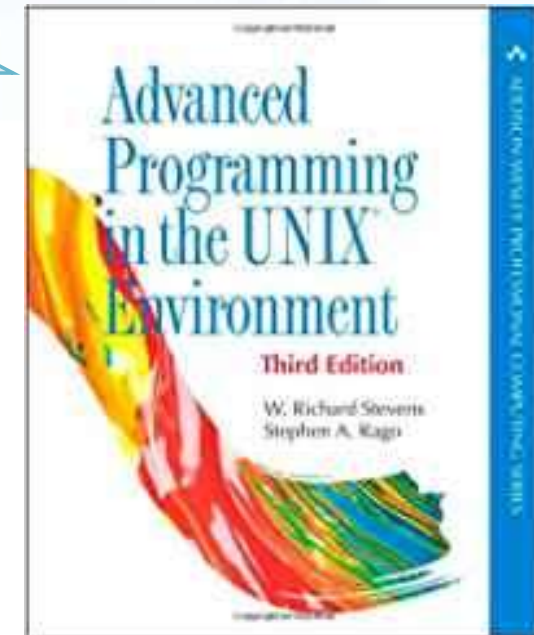
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Premises

❖ Where are we?

- u01-courseIntroduction
- u02-review
- u03-cppBasics
- u04-cppLibrary
- u05-multithreading
- u06-synchronization
- u07-advancedIO
- u08-IPC

Main
reference



- u07s02e
- u07s05e
- u07s01-introduction.pdf
- u07s02-IO.pdf
- u07s03-fileLocking.pdf
- u07s04-multiplexingIO.pdf
- u07s05-fileMapping.pdf

System calls

- ❖ In modern OSs hardware-related services are performed using **system calls**
 - Access a hard disk drive or a device, create and execute a new process, etc.
- ❖ **System calls** are how a program requests a service from the OS
 - They provide an essential interface between a process and the operating system
 - They are well-defined and limited in number
 - Linux and OpenBSD about 300
 - NetBSD and FreeBSD about 500
 - Windows about 2000

For more details, see the material of the OS course and Prof. Cabodi's part

System calls

- ❖ System calls are similar to functions but
 - Require the intervention of the kernel, i.e., a transfer of control from the user space to the kernel space
 - Need some sort of architecture-specific features like a software interrupt or a trap
 - They are by nature slow (as they require the intervention of the kernel)
 - They are usually divided into **slow** and **all the others** system calls

IO system calls

❖ Slow system calls are the ones that can **block** forever

➤ Many are related to IO operations

- **Opens** can block until some condition occurs on certain files
 - Until the unit is attached to the system (e.g., a file)
 - Until another process has the unit open for reading (e.g., a FIFO)
- **Reads** can block if the data is not present
 - Read from stdin or a pipe
- **Writes** can block if they cannot be accepted
 - A write to a full disk or a full pipe a write on a **locked** record in a file

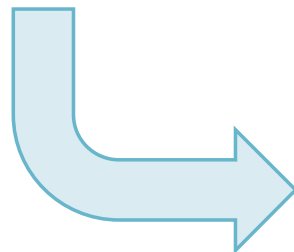
See section u07s03

Synchronous system calls

- ❖ Moreover, standard IO system calls are usually **synchronous**
 - The task waits until the operation completes
 - Synchronous operations are inherently **slow** compared to other processing
 - Delays may be caused by
 - Hardware devices, e.g., track and sector, seek time on random access, etc.
 - Data transfer between a physical device and the system memory, etc.
 - Network transfer between file servers and computing devices, etc.

IO Efficiency

- ❖ In many applications, IO is fundamental and being inefficient implies enormous time penalty
- ❖ The standard UNIX-POSIX introduces numerous advanced IO functionalities such as
 - Non-blocking and asynchronous IO
 - File locking
 - Multiplexing IO
 - Memory-mapped IO



-  u07s02-IO.pdf
-  u07s03-fileLocking.pdf
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-  u07s05-memoryMapping.pdf